1. the name of your game, your names, your section number(s), and “Spring 2026”
Amazooom Delivery Service
Benjamin Church 02
Tatiana Neville 02
Spring 2026

2. at least one image (screenshot) showing a typical scene from your game being played

3. instructions for compiling and running your game, including the network server
Start with clearing both TAGE and game class files
-build TAGE
-compile.bat
-change the ip addresses as needed in run_p1.bat and run_p2.bat
-run_server.bat
-run_p1.bat
-run_p2.bat (optional)

4. how to play your game, including what things happen and how the scoring works

The goal of our game is to deliver 5 packages to houses, avoiding the enemy (helicopter). You start in the “pregame” state, which acts as the character selection. Movement is disabled in this state. Pushing buttons 1 and 2 on the keyboard will change your witch avatar (blue witch, green witch). Once you have made your selection, press the enter key to start the game.

You begin on the ground of the game, with 0 packages on hand. You can move using WASD inside the warehouse, where you can collide with a pile of boxes inside the warehouse. This fills your packages that you have on hand (from 0 to 3).

With your packages refilled, exit the warehouse and press r to fly up in the air. Be careful in the air, as the helicopter will chase you and will end the game if he touches you.

Locate the house with the light above it and a spinning blue arrow, this is the house you must deliver the package to.

Fly above the house, try to get the witch centered above the house as best as you can. When you drop the package, you want the package to hit as close to the center of the house as you can. Press space to drop a package over the house, on success, you will hear a point sound, your points will increment, your packages on hand will decrement, and the blue arrow will move to a different house.

Keep playing, delivering to houses, refilling packages as needed and avoiding the helicopter. You can press r at any time during the game to go up in the air / down to the ground. When you reach 5 points, the HUD will display "You win!". You have the option to continue playing as normal even after you win the game.

5. what player controls are available (what all keyboard/gamepad buttons do, etc.)

- WS & Left Joystick: move forward/back
- AD & Right Joystick Left/Right & Right/Left Arrow Keys: Turn camera right/left
- Right Joystick Up/Down & Arrow Keys Up/Down: Turn camera up/down
- QE: Zoom in/out
- B Button & R: Fly up / drop down
- X Button & Space: Drop a package
- 1 & 2: Select character (only available before "starting game" [pressing enter key])
- A Button & Enter Key: Start game
- ESC: End Game

6. a description of the lighting used in your game, including any lights that turn on/off

- Ambient light (always on)
- One persistent light over the warehouse
- One light over each house. As the game progresses, a house will be "selected" as the "active" house. The light above this house will be turned on. When the "active" house changes, this light turns off, and the light above the new "active" house is turned on (12 lights, one over each house that turn on and off)

7. a brief summary of any changes (or none) that you made to the network protocol

- We expanded the provided networking starter code to support our final witch avatar system. The original networking files only created a simple ghost avatar using one shared ghost shape and texture, and only sent basic position information through create, move, details-for, and bye messages. We modified this system so remote players are represented as witch ghost avatars instead of the original simple ghost object.
- The network protocol now sends the selected avatar/texture value along with create, move, and details-for messages. This allows each client to display the remote player with the correct selected witch texture. Although both avatar options currently use the witch model, the texture choice is still propagated across the network so other players see the correct skin.

- We also added ghost support for the broom. Each ghost avatar creates its own broom as a child object, matching the local player's witch/broom hierarchy. This means remote players appear with both the witch model and the broom model, rather than only a single ghost object.
- We also updated the join/details flow so when a second player joins, the clients request and send details for existing players. This allows the newer client to see players who were already connected. The bye handling was also updated so when a client exits, the corresponding remote ghost avatar can be removed from the other player's game.
- Finally, move-message handling was adjusted so move packets update existing ghosts but do not create new ghosts. This prevents late move packets from recreating a ghost after a player has already exited.

8. a list of changes and additions that you made to TAGE

- RotationController.java: node controller that rotates an object.
- There is also one other node controller, ShakeSinkController.java, but it is unused.

9. a statement indicating the (1) genre, (2) theme, (3) dimensionality, and (4) activities utilized in your game (see week 1 notes -- chapter 00 -- for examples)

- Management
- Fantastical Package Delivery
- 3D
- Exploration, Dodge Enemies, Fly/Walk Witch, Deliver Packages

10. an explanation of where (in the game, not the code) each project requirement is visible

- External Models: Witch Model (UV unwrapped), Broom Model (UV unwrapped), BoxPile Model (UV unwrapped), Mountains (Not UV unwrapped but textured)
- Networked Multiplayer: Multiple players can connect through the network server. Each remote player appears as a ghost witch avatar with the correct selected texture and a broom attached through the scene graph hierarchy. Players can move around and see each other's position update over the network. When a new player joins, the client requests details for existing players so both players can see each other. When a player exits, a bye message is sent so the remote ghost avatar can be removed from the other client.
- Skybox and Terrain: Terrain is implemented (textured with grass and roads) and player moves up/down with the height map when walking/flying, Skybox is implemented and renders behind all objects
- Lights:
    - Ambient light (always on)
    - One persistent light over the warehouse
    - One light over each house. As the game progresses, a house will be "selected" as the "active" house. The light above this house will be turned on. When the "active" house changes, this light turns off, and the light

above the new “active” house is turned on (12 lights, one over each house that turn on and off)

- HUD: Two HUDs, one displays the number of boxes a player has, one displays how many points the player has, as well as win screen, death screen and start screen.
- 3D Sound:
    - Enemy AI has an attached sound that follows then the AI moves (plays on loop)
    - When a package collides with a house scoring the user a point, the box plates an attached“ping” sound
- Hierarchical SceneGraph: The broom is a child of the witch avatar, so translation and rotation propagate from the witch to the broom. This hierarchy is used for both the local player and remote ghost avatars, allowing each remote witch to appear with a matching broom that follows their movement.
- Animation: Witch has a movement animation, hat bounces when the witch moves
- NPCs: An enemy object with a “chase” AI attached, follows the player and ends game for player on collision
- Physics:
    - Player can drop a box, has collision action with heightmap (bounces around) and houses (scores point)
    - Player can collide with BoxPile, refills boxes the player has
- The game is in 3P view, the game has keyboard/gamepad input, the game works in FSEM and windowed modes, uses some additional assets (listed in section 15), game works in both single and multi player modes, all resources use relative paths, IP is configurable in client run.bat program runs on two machines in RVR 5029, package name has been changed

11. a list of the requirements that you weren’t able to get working

- **<u>All requirements for this project have been met.</u>**
- A note should be made that we had intended to do much more with animations, but we had significant issues with getting them to work properly in tage (working properly in blender and exported correctly, but didn’t render properly in tage). Therefore simple animation was added.

12. any technique you used in your game that goes beyond the requirements

- One technique beyond the base requirements was expanding the provided multiplayer ghost system to support hierarchical ghost avatars. The original networking example only displayed a single remote ghost object. In our version, each remote ghost avatar also creates and manages a broom child object, so the remote witch and broom move together through scene graph hierarchy.
- We also expanded the network protocol to carry avatar/texture selection data, allowing remote players to appear with the correct selected witch texture instead of using one fixed ghost texture for all clients.

13. the contributions of each team member, including who designed which model(s)

- Benjamin Church:

    - Main gameplay loop (getting packages, delivering packages, random select house, etc)
    - UV unwrap / Creation of boxPile
    - Creation of broomstick / mountain / arrow models
    - Texturing of heightmap
    - Physics
    - NPC
    - Sound
    - Lights
    - Networking base
    - External Model Texture (Warehouse)
- Tatiana Neville:
    - Base game (built off of A2)
    - Witch model creation
    - Witch model animation
    - Texturing/UV unwrapping Broomstick model
    - Texturing/UV unwrapping Witch (both textures)
    - Hierarchy between broomstick and Witch
    - Networking Ghosting Fixes
    - Skybox
    - Terrain/heightmap setup
    - HUD
    - External Model Texture (Helicopter model, House model)

14. a list of assets that you created yourself (models, textures, heightmap, etc.), and items obtained that were distributed in CSc-155 or CSc-165

- Textures:
    - blue.PNG
    - broomTex.png (UV Unwrap)
    - roads.PNG
    - rock.PNG
    - witchTexB.png and witchTexA.png (UV Unwrap)
    - Boxpile_texture.png (UV Unwrap)
    - heliTex.png (UV Unwrap)
    - houseTex.png (UV Unwrap)
    - Warehouse_texture.png (UV Unwrap)
- Models:
    - arrow.obj
    - boxpile.obj
    - broomModel.obj
    - mountain.obj
    - witchModel.rkm/rks/rka

15. Source and evidence of permission for any item (models, textures, etc.) not listed in #14

- Low Poly House, CC Attribution:
  - https://sketchfab.com/3d-models/low-poly-house-a838c62fbf804d79853faf0cfeb6072e
- Industrial Building 49 PBR 3D model, TurboSquid Standard Model License
  - https://www.turbosquid.com/FullPreview/2532111?dd_referrer=
  - License: https://www.turbosquid.com/licensing
- Helicopter Model, Personal Use License (Not for commercial)
  - https://free3d.com/3d-model/low-poly-helicopter-91325.html

16. which RVR-5029 lab machines (at least two – it's networked!) on which your program was tested and is known to work correctly on

- ECS-PAPERBOY
- ECS-PACMAN